

Efficient Quantum Access Models of Sparse Structured Matrices using Linear Combination of “Things”

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Outline

- Objective and background
- Introduce sigma basis
- Unitary completion concept and circuit construction
- Comparison with related methods
- Applications to variational and fully-fault tolerant quantum algorithms
- Summary

Objective

- Partial differential equations (PDEs)/ordinary differential equations (ODEs) arise in many engineering/science applications.

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- Solutions of PDEs/ODEs, after discretization, can often be cast into system of linear equations

$$\mathbf{Ax} = \mathbf{b}.$$

- $\mathbf{A}$ is typically sparse/structured, e.g., diagonal, block diagonal, banded, Toeplitz:

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & & & 0 \\ -1 & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & -1 \\ 0 & & & -1 & 2 \end{bmatrix}.$$

- Quantum linear system algorithms promise computational advantage
- Efficient quantum data access models are necessary to realize this advantage
- Can the structure/sparsity be leveraged to gain efficiency?

LCU Decomposition: A Versatile Data Access Model

- Given a matrix $\mathbf{A} \in \mathbb{C}^{N \times N}$, where $N = 2^n$, it's linear combination of unitary (LCU) decomposition is defined as

$$\mathbf{A} = \sum_{l=1}^{n_l} \alpha_l \mathbf{U}_l,$$

where $\alpha_l \in \mathbb{C}$, $\mathbf{U}_l \in \mathbb{C}^{2^n \times 2^n}$ are unitary matrices (i.e. $\mathbf{U}_l \mathbf{U}_l^\dagger = \mathbf{U}_l^\dagger \mathbf{U}_l = \mathbf{I}$) and n_l is # LCU terms.

- Most common approach for LCU decomposition uses Pauli basis $\mathbb{P} = \{\mathbf{I}, \sigma_x, \sigma_y, \sigma_z\}$ (or $\mathbf{I}, X, Y, Z$)

$$\begin{aligned}\sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & \sigma_y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, & \mathbf{I} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

so that

$$\mathbf{U}_l = \sigma_0 \otimes \sigma_1 \otimes \cdots \otimes \sigma_{n-1},$$

with $\sigma_i \in \mathbb{P}, i = 0, \dots, n-1$.

- LCU decomposition is a commonly used “data access model” in
 - Variational quantum algorithms (VQA): VQE, VQLS, etc.
 - Fault tolerant algorithms: block encoding based QSP, QVST, etc.

LCU Decomposition in Variational Quantum Algorithms

- Consider the problem of computing ground state of a given Hamiltonian $\mathbf{H}$ (Hermitian, i.e. $\mathbf{H}^\dagger = \mathbf{H}$)

$$\mathbf{H} |\psi_0\rangle = E_0 |\psi_0\rangle$$

- Variational quantum eigensolver (VQE) uses an ansatz state $|\psi(\theta)\rangle$, and classically optimizes the parameters θ

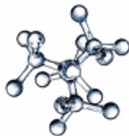
$$\min_{\theta} E(\theta) \equiv \langle \psi(\theta) | \mathbf{H} | \psi(\theta) \rangle \geq E_0$$

- To compute $E(\theta)$, LCU decomposition of $\mathbf{H} = \sum_{i=1}^{n_l} \alpha_i \mathbf{U}_i$ can be used

$$\langle \psi(\theta) | \mathbf{H} | \psi(\theta) \rangle = \sum_{i=1}^{n_l} \alpha_i \langle \psi(\theta) | \mathbf{U}_i | \psi(\theta) \rangle$$

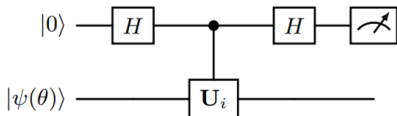
where each term $\langle \psi(\theta) | \mathbf{U}_i | \psi(\theta) \rangle$ can be estimated via the Hadamard test.

- Larger $n_l \Rightarrow$ larger # of Hadamard tests needed



$$\mathcal{H}_{\text{el}} |\Psi_n\rangle = E_n |\Psi_n\rangle$$

$$\mathcal{H}_{\text{el}} = -\sum_i \frac{\nabla_i^2}{m_e} - \sum_I \sum_i \frac{Z_I e^2}{|R_I - r_i|} + \sum_i \sum_{j>i} \frac{e^2}{|r_i - r_j|}.$$



LCU Decomposition in Fault Tolerant Algorithms

- Block encoding of a matrix $\mathbf{A}$ involves

$$\mathbf{U} = \begin{bmatrix} \mathbf{A} & \star \\ \star & \star \end{bmatrix},$$

whose application on an ancilla system of the form $|0\rangle |\psi\rangle$ gives desired action

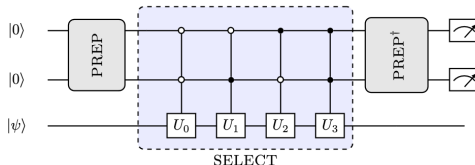
$$\mathbf{U} |0\rangle |\psi\rangle = |0\rangle \mathbf{A} |\psi\rangle + |1\rangle \star |\psi\rangle.$$

- LCU based block encoding can be accomplished by using the PREP and SEL oracles

$$\text{PREP} |0\rangle = \sum_{k=1}^{n_l} \sqrt{\frac{|\alpha_k|}{\lambda}} |k\rangle, \quad \text{SEL} |k\rangle |\psi\rangle = |k\rangle \mathbf{U}_k |\psi\rangle,$$

where $\lambda = \sum_{i=1}^{n_l} |\alpha_i|$ leading to $\langle 0 | \text{PREP}^\dagger \cdot \text{SEL} \cdot \text{PREP} |0\rangle |\psi\rangle = \frac{\mathbf{A} |\psi\rangle}{\lambda}.$

- Larger $n_l \Rightarrow$ increased quantum resources (i.e., #ancillas, #gates and circuit depth) needed for the implementation of the associated circuit



Pauli Basis LCU Decomposition

- $\mathbf{U}_I = \sigma_0 \otimes \sigma_1 \otimes \cdots \otimes \sigma_{n-1}$ with $\sigma_i \in \mathbb{P}, i = 0, \dots, n-1$ can be implemented efficiently on a quantum device.
- In applications, such as quantum chemistry, the Pauli decomposition is often known by construction.
- More generally, numerical techniques have to be employed ¹.
- In general # Pauli LCU terms i.e., n_I can vary as

$$n_I = \begin{cases} \mathcal{O}(2^n), & \text{(diagonal matrix)} \\ \mathcal{O}(2^{2n}), & \text{(dense matrix).} \end{cases}$$

- For sparse matrices that typically arise in solving PDEs/ODEs

$$n_I \text{ can be } > \text{nnz}(\mathbf{A}),$$

where nnz is number of non-zero terms in $\mathbf{A}$.

- Larger $n_I \Rightarrow$ more quantum resource needed for data loading, which can become a key bottleneck in using LCU and impact the quantum advantage in many applications
- By using a different basis we show that poly-logarithmic dependence

$$n_I = \mathcal{O}(\text{poly}(\log N)) = \mathcal{O}(\text{poly}(n)),$$

can be achieved for sparse/structured matrices, where $N = 2^n$ is the matrix size.

¹L. Hantzko, L. Binkowski, and S. Gupta, Tensorized pauli decomposition algorithm, arXiv preprint arXiv:2310.13421, 2023.

Sigma Basis and LCNU Decomposition

- Sigma basis ^{2 3} is a set including non-unitary operators

$$\mathbb{S} = \{\mathbf{I}, \sigma_+, \sigma_-, \sigma_+ \sigma_-, \sigma_- \sigma_+\},$$

where

$$\begin{aligned}\sigma_+ &= |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, & \sigma_- &= |1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \\ \sigma_+ \sigma_- &= |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & \sigma_- \sigma_+ &= |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.\end{aligned}$$

- Given a matrix $\mathbf{A} \in \mathbb{C}^{2^n \times 2^n}$, its **linear combination of non-unitary (LCNU)** decomposition is defined as

$$\mathbf{A} = \sum_{l=1}^{n_l} \alpha_l \mathbf{A}_l,$$

where $\alpha_l \in \mathbb{C}$ and $\mathbf{A}_l = \sigma_0 \otimes \sigma_1 \otimes \cdots \otimes \sigma_{n-1}$ with $\sigma_i \in \mathbb{S}, i = \dots, n-1$.

- Sigma basis offers an efficient decomposition of sparse/structured matrices.
- But this reduction in n_l should not come at the expense of proportionate increase in quantum resources for implementing these non-unitary operators.

² Hai-Ling L., et. al. "Variational quantum algorithm for the Poisson equation." Physical Review A 104.2 (2021): 022418

³ A. Gnanasekaran and A. Surana. "Efficient Variational Quantum Linear Solver for Structured Sparse Matrices." IEEE International Conference on Quantum Computing and Engineering, 2024.

Sigma Basis Decomposition, Motivating Example

- Consider the decomposition of the following sparse matrix in Pauli and sigma basis, respectively

$$\begin{aligned} \text{Pauli: } \begin{bmatrix} & & 1 \\ & 0 & \\ 2 & 0 & \end{bmatrix} &= 0.75 \sigma_x \otimes \sigma_x - 0.25i \sigma_x \otimes \sigma_y - 0.25i \sigma_y \otimes \sigma_x - 0.75 \sigma_y \otimes \sigma_y \\ \text{Sigma: } \begin{bmatrix} & & 1 \\ & 0 & \\ 2 & 0 & \end{bmatrix} &= \sigma_+ \otimes \sigma_+ + 2 \sigma_- \otimes \sigma_- . \end{aligned}$$

- # terms Pauli basis decomposition is 4, whereas sigma basis decomposition require only 2 terms.
- In general, $n_l \leq \text{nnz}(A)$ with sigma basis as desired.
- Next we consider examples of linear/nonlinear PDEs in 1D (results generalize to higher dimensions under some assumptions, see the paper for details).

Sigma Basis Decomposition: Poisson (Elliptic) PDE

- Consider 1D Poisson equation:

$$-\frac{\partial^2 u(x)}{\partial x^2} = f(x), \quad x \in \Omega = [0, l]$$

$$(\text{b.c.}): u(0) = 0, \quad u(l) = 0.$$

- Let $\mathbf{A}\mathbf{u} = \mathbf{b}$ be the linear system generated by central difference discretization scheme, where

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & & & 0 \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ 0 & & & -1 & 2 \end{bmatrix} \in \mathbb{R}^{n_x \times n_x}, \quad (1)$$

is a $n_x \times n_x$ matrix with n_x being # of spatial discretization points and $\mathbf{b} = (\Delta x)^2 (f(x_1), \dots, f(x_{n_x}))^T$.

Sigma Basis Decomposition: Possion PDE

- For simplicity assume $n_x = 2^s$ for some integer $s > 0$.
- We can write a recursive decomposition for $\mathbf{A}$ in sigma basis as follows

$$\mathbf{A} := \mathbf{A}^{(s)} = \begin{bmatrix} \mathbf{A}^{(s-1)} & (\mathbf{D}^{(s-1)})^T \\ \mathbf{D}^{(s-1)} & \mathbf{A}^{(s-1)} \end{bmatrix} = \mathbf{I}_2 \otimes \mathbf{A}^{(s-1)} + \sigma_- \otimes \mathbf{D}^{(s-1)} + \sigma_+ \otimes (\mathbf{D}^{(s-1)})^T,$$

$$\mathbf{D}^{(s-1)} = \begin{bmatrix} 0 & \dots & -1 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} = - \underbrace{\sigma_+ \otimes \dots \otimes \sigma_+}_{s-1 \text{ times}}, \quad \mathbf{A}^{(1)} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 2\mathbf{I}_2 - \sigma_- - \sigma_+.$$

- Thus, $\mathbf{A}$ can be decomposed into $2s + 1 = 2 \log n_x + 1$ terms.

PosLCU_semilog.png

recursion.png

Sigma Basis Decomposition: Heat (Parabolic) PDE

- Consider 1D heat equation:

$$\frac{\partial u(x, t)}{\partial t} = \alpha \frac{\partial^2 u(x, t)}{\partial x^2},$$
$$(\text{b.c.): } -k \frac{\partial u(x, t)}{\partial x} \Big|_{x=0} = q, \quad -k \frac{\partial u(x, t)}{\partial x} \Big|_{x=l} = 0, \quad (\text{i.c.): } u(x, 0) = u_0(x).$$

- Discretize in space/time

$$\left(\mathbf{I}_{n_x} - \frac{\alpha \Delta t}{(\Delta x)^2} \mathbf{A}' \right) \mathbf{u}_{t+1} = \mathbf{u}_t + \frac{q \Delta t}{k \Delta x} \mathbf{e}_1, \quad t = 0, \dots, n_t - 1$$

where, $\mathbf{u}_t = [u_{1,t} \quad u_{2,t} \quad \dots \quad u_{n_x,t}]^T$ and $\mathbf{e}_1 = [1 \quad 0 \dots, 0]^T$.

- Express as a single linear system $\mathbf{A}\mathbf{u} = \mathbf{b}$ where $\mathbf{A}$ is a $n_x n_t \times n_x n_t$ matrix

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_{n_x} & & & 0 \\ -\mathbf{I}_{n_x} & \mathbf{I}_{n_x} & & \\ & \ddots & \ddots & \\ 0 & & -\mathbf{I}_{n_x} & \mathbf{I}_{n_x} \end{bmatrix} - \frac{\alpha \Delta t}{\Delta x^2} \begin{bmatrix} 0 & & & \\ & \mathbf{A}' & & \\ & & \ddots & \\ & & & \mathbf{A}' \end{bmatrix} = \mathbf{A}_1 - \frac{\alpha \Delta t}{\Delta x^2} \mathbf{A}_2.$$
$$\mathbf{u} = [\mathbf{u}_0^T \quad \mathbf{u}_1^T \quad \dots \quad \mathbf{u}_{n_t-1}^T]^T, \quad \mathbf{b} = [\mathbf{u}_0^T \quad \frac{q \Delta t}{k \Delta x} \mathbf{e}_1^T \quad \dots \quad \frac{q \Delta t}{k \Delta x} \mathbf{e}_1^T]^T.$$

Sigma Basis Decomposition: Heat PDE

- For simplicity assume $n_x = 2^s$ and $n_t = 2^t$. Then $\mathbf{A}_1 = \mathbf{A}_1^{(s+t)}$ corresponds to $s + t$ qubits and can be written as

$$\mathbf{A}_1^{(s+t)} = \begin{bmatrix} \mathbf{A}_1^{(s+t-1)} & 0 \\ \mathbf{D}_1^{(s+t-1)} & \mathbf{A}_1^{(s+t-1)} \end{bmatrix}, \mathbf{D}_1^{(s+t-1)} = \begin{bmatrix} 0 & \dots & -\mathbf{I}_{n_x} \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} = -\underbrace{\sigma_+ \otimes \dots \otimes \sigma_+}_{t-1 \text{ times}} \otimes \mathbf{I}_{n_x}$$

- Then following recursive relation holds

$$\mathbf{A}_1^{(s+t)} = \mathbf{I}_2 \otimes \mathbf{A}_1^{(s+t-1)} + \sigma_- \otimes \mathbf{D}_1^{(s+t-1)}, \quad \mathbf{A}_1^{(s+1)} = \begin{bmatrix} \mathbf{I}_{n_x} & 0 \\ -\mathbf{I}_{n_x} & \mathbf{I}_{n_x} \end{bmatrix} = \mathbf{I}_2 \otimes \mathbf{I}_{n_x} - \sigma_- \otimes \mathbf{I}_{n_x}.$$

In sigma basis,

- $\mathbf{A}_1$ can be decomposed into $\log n_t + 1$ terms.
- $\mathbf{A}_2$ can be decomposed into $4 \log n_x + 6$ terms.

HeatLCU_semi-log.png

Sigma Basis Decomposition: Wave (Hyperbolic) PDE

- Consider 1D wave equation:

$$\frac{\partial^2 w(x, t)}{\partial t^2} = c^2 \frac{\partial^2 w(x, t)}{\partial x^2},$$
$$\text{(b.c.): } \left. \frac{\partial w(x, t)}{\partial x} \right|_{x=0} = \left. \frac{\partial w(x, t)}{\partial x} \right|_{x=l} = 0, \quad \text{(i.c.): } w(x, 0) = f_1(x), \quad \left. \frac{\partial w(x, t)}{\partial t} \right|_{t=0} = f_2(x).$$

- Following similar procedure one can show that resulting linear system has $\log n_t + 1 + 2(2 \log 2n_x + 4)$ terms in sigma basis.

WaveLCU_semi-log.png

Sigma Basis Decomposition: Nonlinear PDEs/ODEs

- Consider a nonlinear 1D Burger's equation

$$\begin{aligned}\partial_t u + u \partial_x u &= \nu \partial_x^2 u + f(x, t), \\ u(x, 0) &= u_0(x), u(0, t) = 0, u(L, t) = 0,\end{aligned}$$

with $f(x, t)$ being the forcing function.

- Nonlinear PDEs/ODEs cannot be directly simulated on quantum computers.
- Carleman linearization ⁴ is one popular approach to transform nonlinear ODEs to linear but infinite dimensional system of ODEs which is truncated.

Workflow.pdf

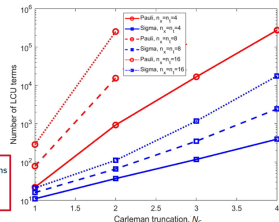
⁴ Jin-Peng Liu et. al. Efficient quantum algorithm for dissipative nonlinear differential equations. Proceedings of the National Academy of Sciences, 118(35), 2021.

Sigma Basis Decomposition: Nonlinear PDEs/ODEs

- Matrix structure with Carleman linearization is complex, yet sigma basis decomposition can be derived analytically ⁵

$$\begin{pmatrix} \mathbf{I} & 0 & 0 & \cdots \\ -\mathbf{I} & [\mathbf{I} - \mathbf{A}_{N_c}(0, \nu)h] & 0 & \cdots \\ 0 & 0 & \ddots & \\ 0 & 0 & -\mathbf{I} & [\mathbf{I} - \mathbf{A}_{N_c}((M-1)h, \nu)h] \end{pmatrix} \begin{pmatrix} \hat{\mathbf{w}}^0 \\ \hat{\mathbf{w}}^1 \\ \vdots \\ \hat{\mathbf{w}}^M \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{b}}(\nu) \\ \hat{\mathbf{w}}(0) \\ hb(0, \nu) \\ \vdots \\ hb((M-1)h, \nu) \end{pmatrix}$$

$$\mathbf{A}_{N_c}(t, \nu) = \begin{pmatrix} \nu \tilde{\mathbf{A}}_1^1 & \mathbf{A}_1^1 & 0 & \cdots & 0 \\ \mathbf{A}_1^1 & \nu \tilde{\mathbf{A}}_2^1 & \mathbf{A}_2^1 & \cdots & 0 \\ 0 & \mathbf{A}_2^1 & \nu \tilde{\mathbf{A}}_3^1 & \mathbf{A}_3^1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \mathbf{A}_{N_c-1}^{N_c} & \nu \tilde{\mathbf{A}}_{N_c}^{N_c} \end{pmatrix} \quad \begin{matrix} \tilde{\mathbf{A}}(\nu) = \tilde{\mathbf{A}}_1 + \nu \tilde{\mathbf{A}}_2 \\ \tilde{\mathbf{A}}_1 : \log n_t + 1 + 2 \left(\frac{2 n_x}{(n_x - 1)^2} \left(n_{N_c}^{N_c} - 1 - N_c(n_x - 1) \right) \right) \text{ terms} \\ \tilde{\mathbf{A}}_2 : 2 (N_c + N_c(N_c + 1) \log n_x) \text{ terms} \end{matrix}$$



- # LCU terms scale poly-logarithmically with n_t (temporal discretization) and polynomially with n_x (spatial discretization) and N_c (Carleman truncation level)- much better than Pauli LCU
- Result was improved ⁶, by including permutation matrices (“things”) in the sigma basis, to obtain poly logarithmic dependence both on n_x and n_t

⁵A. Gnanasekaran, A. Surana, and A. Zhu. "Variational Quantum Framework for Nonlinear Partial Differential Equation Constrained Optimization Using Carleman Linearization", Quantum Information and Computation, 25(3), pp 260-289, 2025.

⁶R. Demirdjjan, H., Thomas and D. Gunlycke. "An Efficient Decomposition of the Carleman Linearized Burgers' Equation." arXiv preprint arXiv:2505.00285, 2025.

Sigma Basis Decomposition: Computational Methods

- Given $\mathbf{A}$, how to determine is LCNU decomposition in sigma basis, i.e.

$$\mathbf{A} = \sum_{l=1}^{n_l} \alpha_l \mathbf{A}_l,$$

where $\mathbf{A}_l = \sigma_1 \otimes \sigma_2 \otimes \cdots \otimes \sigma_n$ with $\sigma_i \in \mathbb{S}, i = 1, \dots, n$.

- For a matrix with single non-zero entry, following theorem gives a simple procedure.

Theorem 1

Let $\mathbf{A}$ be a $2^n \times 2^n$ matrix with a single non-zero entry in position (r, c) $r, c \in \{0, 1, \dots, 2^n - 1\}$. Let the binary representation of $r = \sum_{p=0}^{n-1} 2^p q_r(p)$ and $c = \sum_{p=0}^{n-1} 2^p q_c(p)$. Then, the tensor decomposition of A in sigma basis is given by $\mathbf{A}_s = \mathbf{A}_{rc} (\otimes_p \sigma(q_r(p), q_c(p)))$ where,

$$\sigma(q_r(p), q_c(p)) = \begin{cases} \sigma_+ \sigma_- & \text{if } q_r(p) = 0, q_c(p) = 0 \\ \sigma_+ & \text{if } q_r(p) = 0, q_c(p) = 1 \\ \sigma_- & \text{if } q_r(p) = 1, q_c(p) = 0 \\ \sigma_- \sigma_+ & \text{if } q_r(p) = 1, q_c(p) = 1 \end{cases}.$$

Sigma Basis Decomposition: Computational Methods

- **Numerical:** Express $\mathbf{A}$ as a sum of matrices $\tilde{\mathbf{A}}_i \in \mathbb{C}^{2^n \times 2^n}$ each with a single non-zero entry, i.e.

$$\mathbf{A} = \sum_{i=1}^{\text{nnz}(\mathbf{A})} \tilde{\mathbf{A}}_i,$$

and invoke previous theorem for each $\tilde{\mathbf{A}}_i$.

- With sigma basis decomposition the #terms is always upper-bounded by $\text{nnz}(\mathbf{A})$.
- $\mathbf{I} \in \mathbb{S}$ is not used and this approach may not produce a decomposition with the minimum number of terms.
- **Semi-Analytical:** Mix of analytical and numerical: analytical for recursive/ telescoping matrix structure and numerical for the stay entries

$$\mathbf{A}^{(2)} = \begin{bmatrix} * & * \\ * & * \end{bmatrix} \quad \mathbf{A}^{(4)} = \left[\begin{array}{c|c} \mathbf{A}^{(2)} & * \\ \hline * & \mathbf{A}^{(2)} \end{array} \right] \quad \mathbf{A}^{(8)} = \left[\begin{array}{c|c} \mathbf{A}^{(4)} & \\ \hline * & \mathbf{A}^{(4)} \end{array} \right]$$

Unitary Completion Concept

- Given some elements of $\mathbb{S}$ are non-unitary, how to implement

$$\mathbf{A}_I = \sigma_0 \otimes \sigma_1 \otimes \cdots \otimes \sigma_{n-1} \equiv \otimes_k \sigma_k$$

in form of a quantum circuit?

- There are many options, e.g., unitary dilation, unitary completion⁷ etc.

Definition 1

If $\mathbf{U} : W \rightarrow V$ is a linear operator which preserves inner products, i.e., for any $|w_1\rangle, |w_2\rangle$ in $W \subset V$, $\langle w_1 | \mathbf{U}^* \mathbf{U} | w_2 \rangle = \langle w_1 | w_2 \rangle$, then

- An unitary operator $\bar{\mathbf{U}} : V \rightarrow V$ is the unitary completion of $\mathbf{U}$, if $\bar{\mathbf{U}}$ spans the whole space V and $\bar{\mathbf{U}}|w\rangle = \mathbf{U}|w\rangle \forall |w\rangle \in W$.
- $\mathbf{U}^c := \bar{\mathbf{U}} - \mathbf{U}$ is the orthogonal complement of $\mathbf{U}$.

Such a unitary operator $\bar{\mathbf{U}}$ always exists.

⁷A. Nielsen and I. L. Chuang, Quantum computation and quantum information, Cambridge University Press, 2010

Unitary Completion for Sigma Basis

- The unique unitary completion/complements for sigma basis element

Basis element	Unitary Completion	Orthogonal Complement
σ_+	σ_x	σ_-
σ_-	σ_x	σ_+
$\sigma_+\sigma_-$	I	$\sigma_-\sigma_+$
$\sigma_-\sigma_+$	I	$\sigma_+\sigma_-$
I	I	0

- Let $\mathbf{A}_I = \otimes_k \sigma_k$, define

$$\bar{\mathbf{A}}_I := \otimes_k \bar{\sigma}_k, \quad \bar{\sigma}_k \in \{\sigma_x, \mathbf{I}\} \text{ is unitary completion of } \sigma_k \quad (2)$$

$$\mathbf{A}_I^c := \bar{\mathbf{A}}_I - \mathbf{A}_I.$$

- Then it can be proved that

- $\bar{\mathbf{A}}_I$ is unitary completion of $\mathbf{A}_I$.
- $\mathbf{A}_I^c$ is orthogonal complement of $\mathbf{A}_I$.
- $\mathbf{A}_I^T \mathbf{A}_I^c = 0$
- $\mathbf{A}_I^T \mathbf{A}_I + (\mathbf{A}_I^c)^T \mathbf{A}_I^c = \mathbf{I}$

Unitary Completion for Sigma Basis

- Consider an operator $\mathbf{U}_I$ associated with operator $\mathbf{A}_I$ of the form

$$\mathbf{U}_I = \begin{bmatrix} \mathbf{A}_I & \mathbf{A}_I^c \\ \mathbf{A}_I^c & \mathbf{A}_I \end{bmatrix},$$

then, $\mathbf{U}_I$ is a unitary operator whose application on an ancilla system $|0\rangle |\psi\rangle$ leads to

$$\mathbf{U}_I |0\rangle |\psi\rangle = |0\rangle \mathbf{A}_I |\psi\rangle + |1\rangle \mathbf{A}_I^c |\psi\rangle.$$

- It is tedious to work with $\mathbf{A}_I^c$ (as it does not have closed form tensor product expression), so we equivalently express $\mathbf{U}_I$ as $\mathbf{U}_I = \mathbf{U}_{I,1} \mathbf{U}_{I,2}$ with

$$\mathbf{U}_{I,1} = \begin{bmatrix} \mathbf{I} - \mathbf{A}_I \mathbf{A}_I^T & \mathbf{A}_I \mathbf{A}_I^T \\ \mathbf{A}_I \mathbf{A}_I^T & \mathbf{I} - \mathbf{A}_I \mathbf{A}_I^T \end{bmatrix}, \quad \mathbf{U}_{I,2} = \sigma_x \otimes \bar{\mathbf{A}}_I.$$

- Note that $\mathbf{A}_I \mathbf{A}_I^T$ is a binary diagonal matrix with 1's and 0's at the diagonal, since

$$\mathbf{A}_I \mathbf{A}_I^T = \otimes_p \sigma_p \sigma_p^T, \quad \sigma_p \sigma_p^T \in \{\sigma_+ \sigma_-, \sigma_- \sigma_+, \mathbf{I}\}.$$

Therefore, $\mathbf{U}_{I,1}$ is a permutation matrix of size $2^{n+1} \times 2^{n+1}$, which can be implemented using only Toffoli gates.

- $\mathbf{U}_{I,2}$ only involves tensor products of $\sigma_x, \mathbf{I}$, and thus can be implemented efficiently using the single qubit gates.

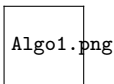
n -controlled Toffoli gate

We use C^nX to represent the n -controlled Toffoli gate:

- C^nX gate acts on $n + 1$ qubits and swaps two rows whose binary representation differ in the k -th bit,
 - with k -th bit being the target qubit, and
 - open (closed) control at $p \neq k$ -th qubit implies 0(1) bit at the p -th location in the binary representation of the row index.
- C^1X is the CNOT gate and C^2X is the Toffoli gate or CCNOT gate.

Toffoli.png

Circuit Implementation of U_I : Pseudo Code and Example

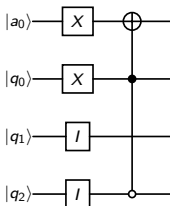


Just needs 5 simple steps!

Consider a $n = 3$ qubit system on q_0, q_1, q_2 , with

$$\mathbf{A}_I = \sigma_- \otimes \mathbf{I} \otimes \sigma_+ \sigma_-.$$

- 1 Compute the unitary completion of $\mathbf{A}_I$ as $\bar{\mathbf{A}}_I = \bar{\sigma}_- \otimes \bar{\mathbf{I}} \otimes \bar{\sigma}_+ \bar{\sigma}_- = \sigma_x \otimes \mathbf{I} \otimes \mathbf{I}$.
- 2 Construct the single-qubit gates X on q_0 and $\mathbf{I}$ on q_1, q_2 .
- 3 Apply X gate on a_0 (recall $\mathbf{U}_{I,2} = \sigma_x \otimes \bar{\mathbf{A}}_I$).
- 4 Compute k as the number of factors in representation of $\mathbf{A}_I$ that are Identity operations. In this case, $k = 1$ as the second term in the tensor product is $\mathbf{I}$.
- 5 Construct a $C^{n-k}X = C^2X$ gate (CCNOT gate) by adding a closed control on q_0 , open control on q_2 and a X gate on target a_0 .



Correctness of Circuit Implementation of U_I

Theorem 2

U_I can be implemented using at most $n + 1$ single qubit gates and a $C^m X$ gate where $m \leq n$.

Lemma 1

A $C^n X$ gate corresponds to a permutation matrix of size $2^{n+1} \times 2^{n+1}$ that permutes two rows whose binary representation differ by a single bit.

Corollary 1

$2k + 1$ $C^n X$ gates are needed to permute two rows in a $2^{n+1} \times 2^{n+1}$ matrix that differ by $k + 1$ bits in their binary representation.

Lemma 2

If $\mathbf{A}_I = \otimes \sigma_p$, where $\sigma_p \in \mathbb{S}$, $r = \sum_{p=0}^{n-1} 2^{n-1-p} q_r(p)$, and $c = \sum_{p=0}^{n-1} 2^{n-1-p} q_c(p)$, then

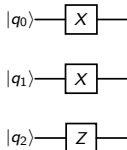
$$\mathbf{A}_I(r, c) = 1 \iff \sigma_p(q_r(p), q_c(p)) = 1 \quad \forall p,$$

where $q_r(p), q_c(p) \in \{0, 1\}$.

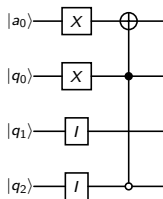
Comparison with Pauli Basis LCU

- $C^n X$ gate can be efficiently implemented using single qubit gates and CNOT gates resulting in a circuit of depth atmost $\mathcal{O}(\log n)$ and size $\mathcal{O}(n)$ with 1 ancilla qubit ⁸.
- Sigma basis requires $C^m X$ gate where $m \leq \log N$.
- So total two ancillas needed: 1 for unitary completion and 1 for implementing $C^m X$ gate.

Method	# LCU terms	#Ancillas qubits	#Gates	Circuit depth
Pauli	$\mathcal{O}(N) - \mathcal{O}(N^2)$ (diagonal-dense)	0	$\log N$	1
Sigma	$\mathcal{O}(\log N)$	2	$\log N + \mathcal{O}(m)$	$1 + \mathcal{O}(\log m)$



Pauli: $\mathbf{A}_I = X \otimes X \otimes Z$



Sigma: $\mathbf{A}_I = \sigma_- \otimes \mathbf{I} \otimes \sigma_+ \sigma_-$

⁸Nie, J., Zi, W., and Sun, X. (2024). Quantum circuit for multi-qubit Toffoli gate with optimal resource. arXiv preprint arXiv:2402.05053.

Comparison with Unitary Dilation

Definition 2

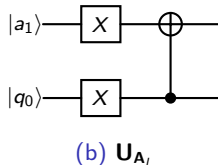
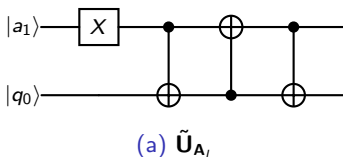
Let $\mathbf{A}$ be contraction, i.e., $\|\mathbf{A}\|_2 \leq 1$, then the unitary dilation $\tilde{\mathbf{U}}_{\mathbf{A}}$ of $\mathbf{A}$ is defined with dilation operator $\mathbf{D}_{\mathbf{A}} = \sqrt{\mathbf{I} - \mathbf{A}^\dagger \mathbf{A}}$ as:

$$\tilde{\mathbf{U}}_{\mathbf{A}} = \begin{pmatrix} \mathbf{A} & \mathbf{D}_{\mathbf{A}} \\ \mathbf{D}_{\mathbf{A}}^\dagger & -\mathbf{A}^\dagger \end{pmatrix},$$

- For $\mathbf{A}_I$ given as tensor product over sigma basis, $(\mathbf{I} - \mathbf{A}_I^\dagger \mathbf{A}_I)^2 = \mathbf{I} - \mathbf{A}_I^T \mathbf{A}_I$, and thus

$$\tilde{\mathbf{U}}_{\mathbf{A}_I} = \begin{bmatrix} \mathbf{A}_I & \mathbf{I} - \mathbf{A}_I \mathbf{A}_I^T \\ \mathbf{I} - \mathbf{A}_I^T \mathbf{A}_I & -\mathbf{A}_I^T \end{bmatrix} \sim \begin{bmatrix} \mathbf{A}_I & \mathbf{I} - \mathbf{A}_I \mathbf{A}_I^T \\ \mathbf{I} - \mathbf{A}_I^T \mathbf{A}_I & \mathbf{A}_I^T \end{bmatrix}.$$

- In general, $\tilde{\mathbf{U}}_{\mathbf{A}_I}$ requires more gates than $\mathbf{U}_I$ and hence, is less efficient, e.g., $\mathbf{A}_I = \sigma_-$



- Unitary completion is however more restrictive in when it can be applied.

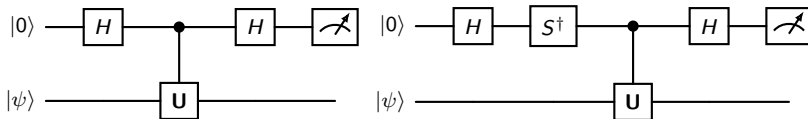
Application: VQA's with Sigma Basis

- Given the LCNU decomposition of $\mathbf{A}$, VQAs often require computing terms of the form

$$\langle \psi_1 | \mathbf{A}_I | \psi_2 \rangle, \quad \langle \psi_1 | \mathbf{A}_i^* \mathbf{M} \mathbf{A}_j | \psi_2 \rangle$$

where $|\psi_1\rangle$ and $|\psi_2\rangle$ are arbitrary states and $\mathbf{M}$ is a unitary matrix.

- Using the Hadamard test combined with the approach for constructing unitary completion $\mathbf{U}_I$, one can design efficient circuit to compute terms of above form



(a) $\text{Re}(\langle \psi | \mathbf{U} | \psi \rangle)$

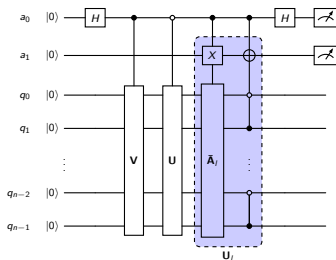
(b) $\text{Im}(\langle \psi | \mathbf{U} | \psi \rangle)$

VQA: Evaluation of $\langle \psi_1 | \mathbf{A}_I | \psi_2 \rangle$

- Real part can be computed by measuring the two ancillas and computing

$$P_{00} - P_{10} = \frac{1}{4} \left(\| |\psi_1\rangle + \mathbf{A}_I |\psi_2\rangle \|^2 - \| |\psi_1\rangle - \mathbf{A}_I |\psi_2\rangle \|^2 \right) = \text{Re} \langle \psi_1 | \mathbf{A}_I | \psi_2 \rangle = \text{Re} \langle 0 | \mathbf{U}^* \mathbf{A}_I \mathbf{V} | 0 \rangle$$

- Imaginary part can be computed by inserting the phase gate on a_0 .

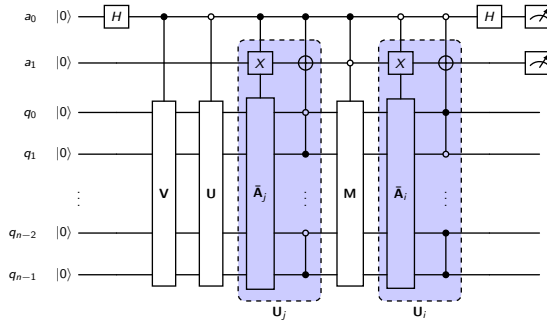


$$(|\psi_1\rangle = \mathbf{U} |0\rangle, |\psi_2\rangle = \mathbf{V} |0\rangle)$$

$$\begin{aligned} |0^{n+2}\rangle &\xrightarrow{H_{a_0}, O_{C_U}, C_V} \frac{1}{\sqrt{2}} (|00\rangle |\psi_1\rangle + |10\rangle |\psi_2\rangle) \xrightarrow{C_{U_I}} \frac{1}{\sqrt{2}} (|00\rangle |\psi_1\rangle + |1\rangle \mathbf{U}_I |0\rangle |\psi_2\rangle) \\ &= \frac{1}{\sqrt{2}} (|00\rangle |\psi_1\rangle + |10\rangle \mathbf{A}_I |\psi_2\rangle + |11\rangle \mathbf{A}_I^c |\psi_2\rangle) \\ &\xrightarrow{H_{a_0}} \frac{1}{2} \left(|00\rangle (|\psi_1\rangle + \mathbf{A}_I |\psi_2\rangle) + |10\rangle (|\psi_1\rangle - \mathbf{A}_I |\psi_2\rangle) + |01\rangle \mathbf{A}_I^c |\psi_2\rangle - |11\rangle \mathbf{A}_I^c |\psi_2\rangle \right). \end{aligned}$$

VQA: Evaluation of $\langle 0 | \mathbf{U}^* \mathbf{A}_i^* \mathbf{M} \mathbf{A}_j \mathbf{V} | 0 \rangle$

- Follow similar procedure as in the previous slide.



$$(|\psi_1\rangle = \mathbf{U} |0\rangle, |\psi_2\rangle = \mathbf{V} |0\rangle)$$

Application: Variational Quantum Linear Solver (VQLS)

VQLS is a hybrid classical quantum algorithm for solving linear systems of the form ⁹:

$$\mathbf{A}|\mathbf{x}\rangle = |\mathbf{b}\rangle, \quad |\mathbf{x}\rangle, |\mathbf{b}\rangle \in \mathbb{R}^N, \mathbf{A} \in \mathbb{R}^{N \times N}, N = 2^n$$

- Let $\mathbf{U}$ be a unitary such that $|\mathbf{b}\rangle = \mathbf{U}|\mathbf{0}\rangle$.
- Choose ansatz $\mathbf{V}(\theta)$, such that $|\psi\rangle = \mathbf{V}(\theta)|\mathbf{0}\rangle$.
- Select cost function:
 - Global cost function:

$$C^{ug}(\theta) = \langle\psi|H^g|\psi\rangle, \quad C^g = \frac{C^{ug}}{\langle\phi|\phi\rangle}, \quad H^g = \mathbf{A}^*\mathbf{U}(\mathbf{I} - |\mathbf{0}\rangle\langle\mathbf{0}|)\mathbf{U}^*\mathbf{A},$$

where, $|\phi\rangle = \mathbf{A}|\psi\rangle$.

- Local cost function (impacted less by barren plateaus)

$$C^{ul} = \langle\psi|H^l|\psi\rangle, \quad C^l = \frac{C^{ul}}{\langle\phi|\phi\rangle}, \quad H^l = \mathbf{A}^*\mathbf{U}\left(\mathbf{I} - \frac{1}{n}\sum_{k=1}^n|0_k\rangle\langle 0_k| \otimes \mathbf{I}_{\tilde{k}}\right)\mathbf{U}^*\mathbf{A}.$$

- Build quantum circuits to compute global/local cost function on a quantum device and classically optimize θ .

⁹Bravo-Prieto, Carlos, et al. "Variational quantum linear solver." Quantum 7 (2023): 1188.

Application: VQLS Cost Function Evaluation

- Compute sigma basis LCNU decomposition of $\mathbf{A}$ using numerical/semi-analytical approach

$$\mathbf{A} = \sum_{l=1}^{n_l} \alpha_l \mathbf{A}_l.$$

- Global cost function:

$$C^g = 1 - \frac{|\langle \mathbf{b} | \phi \rangle|^2}{\langle \phi | \phi \rangle}, \quad \langle \phi | \phi \rangle = \sum_{i,j=1}^{n_l} \beta_{ij} \alpha_i \alpha_j^*, \quad |\langle \mathbf{b} | \phi \rangle|^2 = \sum_{i,j=1}^{n_l} \gamma_{ij} \alpha_i \alpha_j^*,$$
$$\beta_{ij} = \langle \mathbf{0} | \mathbf{V}^* \mathbf{A}_j^* \mathbf{A}_i \mathbf{V} | \mathbf{0} \rangle, \quad \gamma_{ij} = \langle \mathbf{0} | \mathbf{U}^* \mathbf{A}_i \mathbf{V} | \mathbf{0} \rangle \langle \mathbf{0} | \mathbf{V}^* \mathbf{A}_j^* \mathbf{U} | \mathbf{0} \rangle.$$

- Local cost function:

$$C^l = 1 - \frac{1}{n} \frac{1}{\langle \phi | \phi \rangle} \sum_{k=1}^n \sum_{i,j=1}^{n_l} \frac{\alpha_i \alpha_j^*}{2} (\beta_{ij} + \delta_{ijk}), \quad \delta_{ijk} = \langle \mathbf{0} | \mathbf{V}^* \mathbf{A}_j^* \mathbf{U} (\mathbf{Z}_k \otimes \mathbf{I}_{\tilde{k}}) \mathbf{U}^* \mathbf{A}_i \mathbf{V} | \mathbf{0} \rangle.$$

- Use Hadamard like test for evaluating γ_{ij} , β_{ij} and δ_{ijk} .
- Need $O(n_l^2)/O(nn_l^2)$ Hadamard tests for computing global/local cost function.

Application: PDE Constrained Optimization

- Developed a novel hybrid classical quantum framework ^{10 11}
 - Efficient mapping of the optimization problem into a quantum variational form,
 - PDE simulation (computational bottleneck) on a quantum device via VQLS,
 - Design optimization on a classical device via Bayesian Optimization (BO).

WorkflowSchema.pdf

¹⁰ A. Surana, and A. Gnanasekaran. "Variational Quantum Framework for Linear Partial Differential Equation Constrained Optimization." <https://arxiv.org/pdf/2405.16651>, 2024

¹¹ A. Gnanasekaran, A. Surana, and A. Zhu. "Variational Quantum Framework for Nonlinear Partial Differential Equation Constrained Optimization Using Carleman Linearization", Quantum Information and Computation, 25(3), pp 260-289, 2025.

Application: PDE Constrained Optimization

- Pauli LCU decomposition became a key bottleneck in our implementation.
- Use of sigma basis significantly sped up the computation:
 - Heat transfer optimization: VQLS solution time ~ 6 min (sigma), 72 min (Pauli)
 - Inverse problem: Pauli basis intractable, # Pauli basis terms orders of magnitude larger than # sigma basis terms.

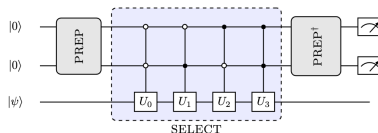


PDEOptNew.png

Application: Block Encoding with Sigma Basis

- Recall PREP and SEL oracles used for LCU based block encoding

$$\text{PREP } |0\rangle = \sum_{k=1}^{n_l} \sqrt{\frac{|\alpha_k|}{\lambda}} |k\rangle, \quad \text{SEL } |k\rangle |\psi\rangle = |k\rangle \mathbf{U}_k |\psi\rangle,$$



- For Pauli LCU the PREP and SEL oracles can be implemented with following resources (Clifford+T complexity) ¹² estimates:

Protocol	n_{anc}	Count	Depth
PREP	$\Omega(\log n_l) \leq n_{\text{anc}} \leq \mathcal{O}(n_l)$	$\mathcal{O}(n_l \log 1/\epsilon)$	$\tilde{\mathcal{O}}(n_l \log(1/\epsilon) \frac{\log n_{\text{anc}}}{n_{\text{anc}}})$
SEL	$\Omega(\log n_l + \log N) \leq n_{\text{anc}} \leq \mathcal{O}(n_l \log N)$	$\mathcal{O}(n_l \log N)$	$\mathcal{O}(n_l \log N \frac{\log n_{\text{anc}}}{n_{\text{anc}}})$

where, n_{anc} denote #ancilla qubits, ϵ is accuracy in implementing PREP and $\tilde{\mathcal{O}}$ hides doubly logarithmic factors.

- Note linear dependence on n_l .

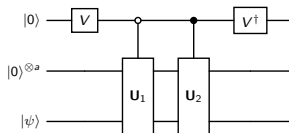
¹²X. M Zhang. and X. Yuan, "Circuit complexity of quantum access models for encoding classical data", npj Quantum Information, 10(1), 42, 2024.

Application: Block Encoding with Sigma Basis

- With the sigma basis, the PREP and SEL oracles can be analogously defined as

$$\text{PREP } |0\rangle = \sum_{k=1}^{n_I} \sqrt{\frac{|\alpha_k|}{\lambda}} |k\rangle, \quad \text{SEL } |k\rangle |\psi\rangle = |k\rangle \mathbf{A}_k |\psi\rangle,$$

- Block-encodings have extensibility properties, i.e., if $\mathbf{U}_1$ and $\mathbf{U}_2$ are block encodings of $\mathbf{A}_1$ and $\mathbf{A}_2$, respectively, then the circuit



is a block-encoding of $|V_{00}|^2 \mathbf{A}_1 + |V_{01}|^2 \mathbf{A}_2$.

- Applying this linear combination of block encoding result, one can obtain sigma basis-based block encoding.
- Resource estimates: Assuming

$$\begin{aligned} \text{Pauli: } n_I &\sim \mathcal{O}(N), \\ \text{Sigma: } n_I &\sim \mathcal{O}(\text{poly}(\log N)), \end{aligned}$$

and using similar analysis as in previous slide, one can show that sigma basis provides an exponential reduction in gate count and circuit depth compared to the Pauli basis.

Summary

- Explored non-unitary sigma basis for LCU type decomposition which is key data access model in many quantum algorithms.
- Presented efficient numerical/semi-analytical techniques for computing the sigma basis decomposition.
- Constructed efficient quantum circuits using the concept of unitary completion.
- Demonstrated **exponential reduction** in the # of decomposition terms for structured/sparse matrices (e.g. arising from PDEs/ODEs discretization). This comes at a **slightly** increased cost of:
 - 1 additional qubit, $O(\log N)$ additional gates,
 - $O(\log \log N)$ increase in circuit depth,compared to the Pauli basis LCU.
- Significant implications for both variational and fully fault tolerant quantum algorithms.
- Provides a general paradigm ("**linear combination of things**") for constructing application-specific efficient decompositions .

Acknowledgement

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